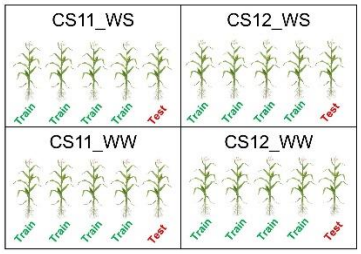
a)

Across-environment prediction: CV2 and CV1

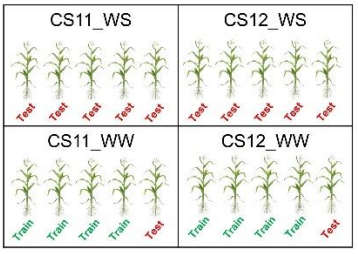


	CV2	CV1
Training Data	4/5 folds from 4 env.	4/5 folds from 4 env.
Test Data	4/5 folds from 4 env.	1/5 folds from 4 env.

Predicting known (CV2) and unknown (CV1) hybrids in characterized environments

b)

Across-environment prediction: CV0 and CV00

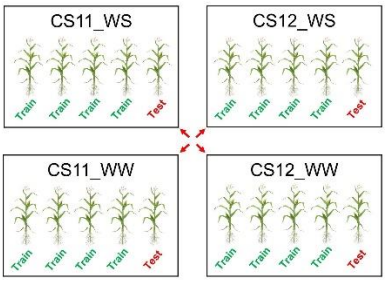


	CV0	CV00
Training Data	4/5 folds from 2 env.	4/5 folds from 2 env.
Test Data	4/5 folds from 2 env.	1/5 folds from 2 env.

Predicting known (CV0) and unknown (CV00) hybrids in uncharacterized environments

c)

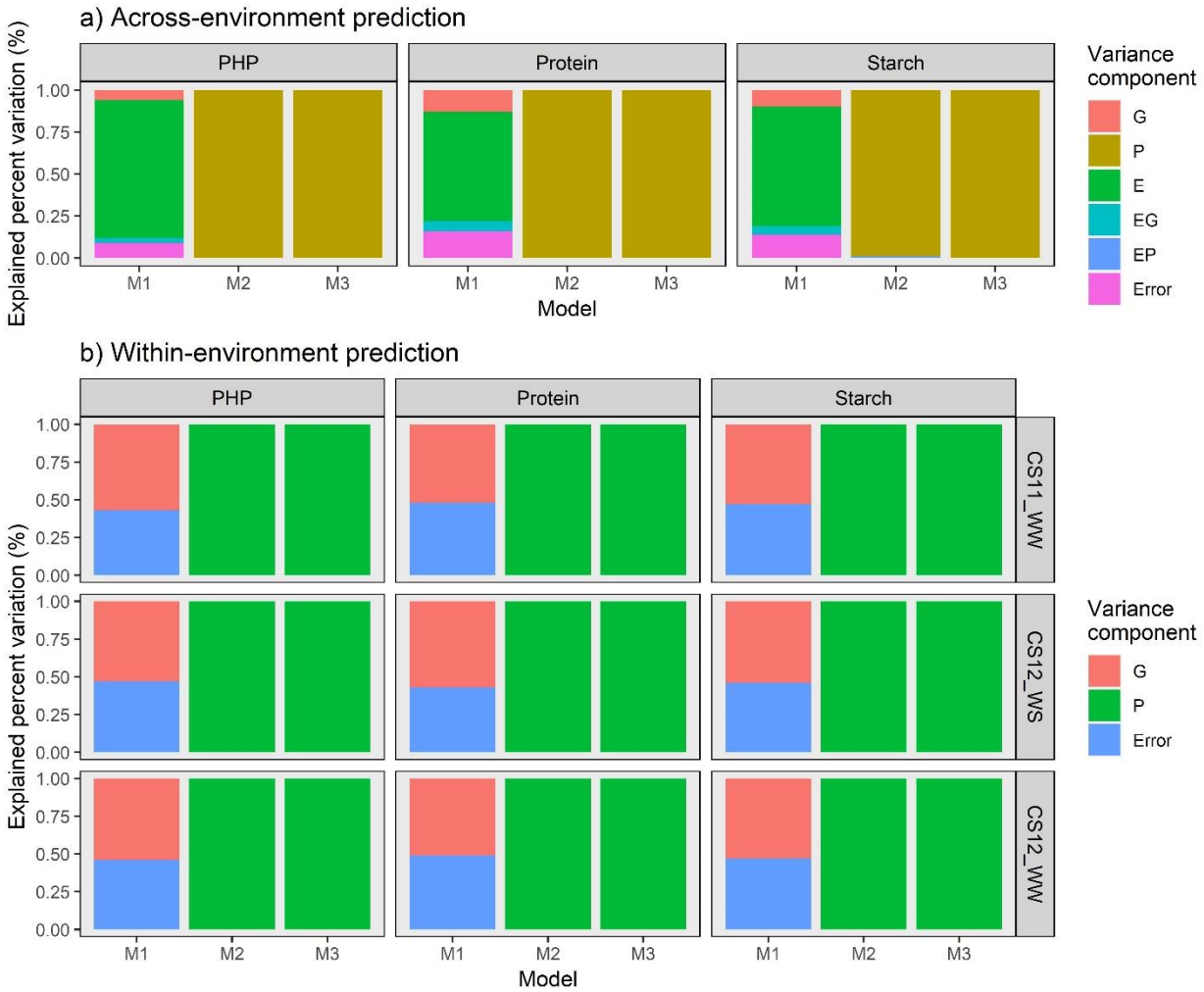
Within-environment prediction: CV1-W



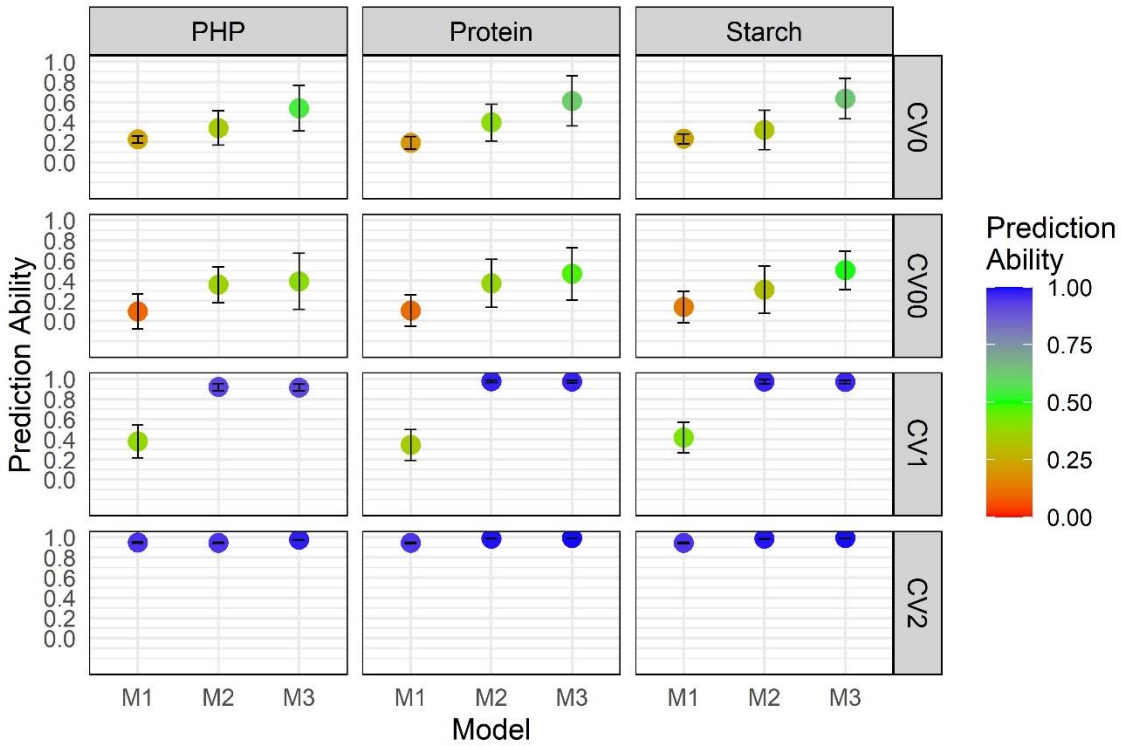
	CV1
Training Data	4/5 folds from single env.
Test Data	1/5 folds from single env. (same env. as training data)

Predicting unknown hybrids in characterized <u>single</u> environments (CV1)

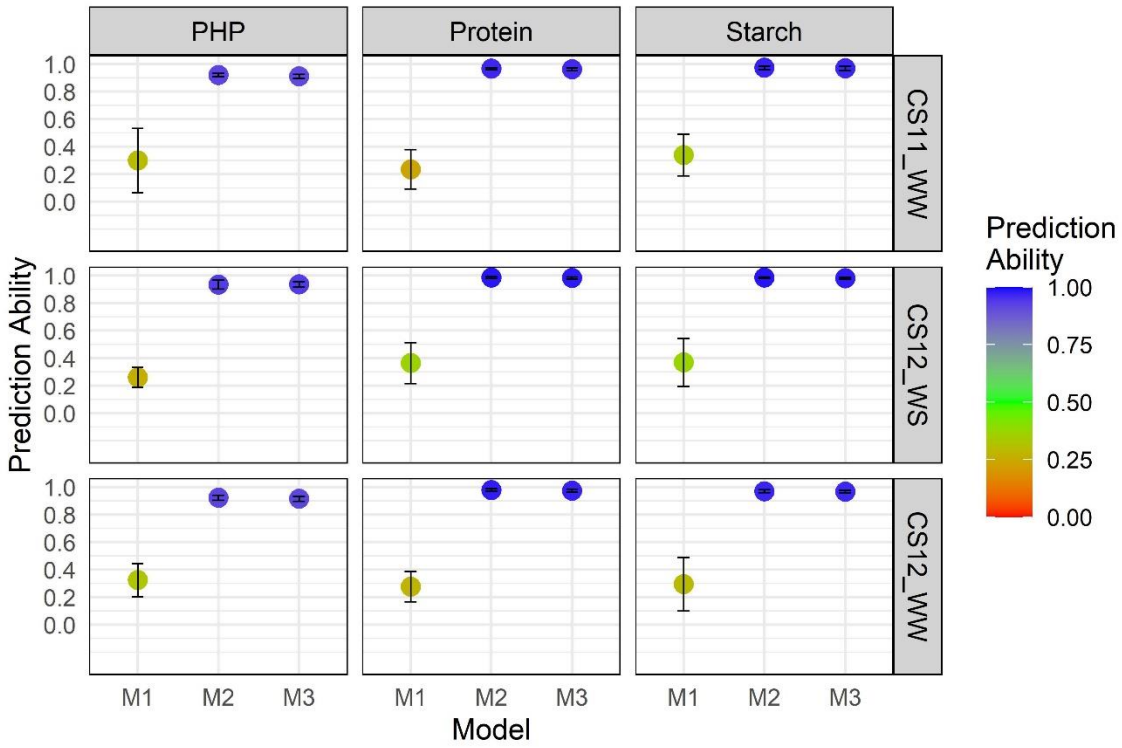
Supplementary Figure 1. **A)** Illustration of across-environment cross-validation. CV2 refers to prediction of known hybrids in known environments while CV1 refers to prediction of unknown hybrids in known environments. **B)** CV0 refers to prediction of unknown hybrids in known environments while CV00 refers to prediction of unknown hybrids in unknown environments. **C)** Illustration of within-environment cross-validation. CV1-W refers to prediction of unknown hybrids in known single environments.



Supplementary Figure 2. Explained percent variation of genomic (M1), phenomic (M2), and combined genomic/phenomic prediction (M3) model components is displayed for (**a**) across-environment prediction and (**b**) within-environment prediction, where G: genomic kernel; E: environment kernel; EG: interaction kernel between genomic and environment kernels; EP: interaction kernel between phenomic and environment kernels; Error: unexplained (residual) variation. Data were not recorded in CS11_WS due to extreme drought conditions.



Supplementary Figure 3. Prediction abilities of across-environment genomic (M1), phenomic (M2), and combined genomic/phenomic prediction (M3) are displayed for kernel composition traits stratified by trait and cross-validation scheme. Prediction abilities close to 1 are indicated in dark blue while those closer to 0 are shown in red. Model notation is described in detail in **Table 2**. M1: environment (E), genomic (G), and environment×genomic (EG) prediction kernels; M2: E, phenomic (P), and environment×phenomic (EP) prediction kernels; M3: E, G, P, EG, and EP prediction kernels.



Supplementary Figure 4. Prediction abilities of within-environment genomic prediction (M1) are reported for kernel composition traits stratified by trait and environment. Only one cross-validation scheme was implemented for within-environment prediction (prediction of unknown genotypes in characterized environments, CV1-W). Prediction abilities close to 1 are indicated by a dark blue color while those closer to 0 appear red. Model notation is described in **Table 3**. M1: genomic (G) prediction kernel; M2: phenomic (P) prediction kernel; M3: G and P prediction kernels.